

Developing a reporting guideline for social and psychological intervention trials

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Abstract

Objectives Social and psychological interventions are often complex. Understanding randomised controlled trials (RCTs) of these complex interventions requires a detailed description of the interventions tested and the methods used to evaluate them;

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however, RCT reports often omit, or inadequately report, this information. Incomplete and inaccurate reporting hinders the optimal use of research, wastes resources, and fails to meet ethical obligations to research participants and consumers.

Methods In this paper, we explain how reporting guidelines have improved the quality of reports in medicine, and describe the on-going development of a new reporting guideline for RCTs: CONSORT-SPI (an extension for social and psychological interventions).

Results and conclusions We invite readers to participate in the project by visiting our website, in order to help us reach the best-informed consensus on these guidelines (<http://tinyurl.com/CONSORT-study>).

Keywords Randomised controlled trial · RCT · CONSORT-SPI · Reporting guideline · Reporting standards

Introduction

Social and psychological interventions aim to improve physical health, mental health, and associated social outcomes. They are often complex and typically involve multiple, interacting intervention components (e.g. several behaviour change techniques) that may act and target outcomes on several levels (e.g. individual, family, community) (Medical Research Council 2008). Moreover, these interventions may be contextually dependent upon the hard-to-control environments in which they are delivered (e.g. health care settings, correctional facilities) (Bonell 2002; Pawson et al. 2004). The functions and processes of these interventions may be designed to accommodate particular individuals or contexts, taking on different forms while still aiming to achieve the same objective (Bonell et al. 2012; Hawe et al. 2004).

Complex interventions are common in public health, psychology, education, social work, criminology, and related disciplines. For example, Multisystemic Therapy (MST) is an intensive intervention for juvenile offenders. Based on social ecological and family systems theories, MST providers target a variety of individual, family, school, peer, neighbourhood, and community influences on psychosocial and behavioural problems (Henggeler et al. 2002). Treatment teams of professional therapists and caseworkers work with individuals, their families, and their peer groups to provide tailored services (Littell et al. 2009). These services may be delivered in homes, social care, and community settings. Other examples of social and psychological interventions may be found in reviews by the Cochrane Collaboration (2013) (e.g. the Developmental, Psychosocial, and Learning Problems Group; the Cochrane Public Health Group) and the Campbell Collaboration (2013).

To understand their effects and to keep services up to date, academics, policymakers, journalists, clinicians, and consumers rely on research reports of intervention studies in scientific journals. Such reports should explain the methods, including the design, delivery, uptake, and context of interventions, as well as subsequent results. Accurate, complete, and transparent reporting is essential for readers to make best use of new evidence, to achieve returns on research investment, to meet ethical obligations to research participants and consumers of interventions,

and to minimise waste in research. However, reports of randomised controlled trials (RCTs) are often poorly reported within and across disciplines including criminology (Perry et al. 2010), social work (Naleppa and Cagle 2010), education (Torgerson et al. 2005), psychology (Michie et al. 2011; Stinson et al. 2003), and public health (Semaan et al. 2002). Biomedical researchers have developed guidelines to improve the reporting of RCTs of health-related interventions (Schulz et al. 2010). However, many social and behavioural scientists have not fully adopted these guidelines, which may not be wholly adequate for social and psychological interventions in their current form (Bonell et al. 2006; Davidson et al. 2003; Perry et al. 2010; Stinson et al. 2003). Because of the unique features of these interventions, updated reporting guidance is needed.

This article describes the development of a reporting guideline that aims to improve the quality of reports of RCTs of social and psychological interventions. We explain how reporting guidelines have improved the quality of reports in medicine, and why guidelines have not yet improved the quality of reports in other disciplines. We then introduce a plan to develop a new reporting guideline for RCTs—CONSORT-SPI (an extension for social and psychological interventions)—which will be written using recommended techniques for guideline development and dissemination (Moher et al. 2010a, b). Wide stakeholder involvement and consensus are needed to create useful, acceptable, and evidence-based guidelines, so we hope to recruit stakeholders from multiple disciplines and professions.

Randomised trials are not the only rigorous method for evaluating interventions; many alternatives exist when RCTs are not possible or appropriate due to scientific, practical, and ethical concerns (Bonell et al. 2011). Nonetheless, RCTs are important to policymakers, practitioners, scientists, and service users, as they are generally considered the most valid and reliable research method for estimating the effectiveness of interventions (Chalmers 2003). Moreover, many of the issues faced in reporting RCTs also relate to other evaluation designs. As a result, this project will focus on standards for RCTs, which could then also inform the development of future guidelines for other evaluation designs.

Impact of CONSORT guidelines

Reporting guidelines list (in the form of a checklist) the minimum information required to understand the methods and results of studies. They do not prescribe research conduct, but facilitate the writing of transparent reports by authors and appraisal of reports by research consumers. For example, the Consolidated Standards of Reporting Trials (CONSORT) Statement 2010 is an evidence-based guideline; to identify items, the developers reviewed evidence of trial design and conduct that could contribute to bias. Using consensus methods, they developed a checklist of 25 items and a flow diagram (Schulz et al. 2010). CONSORT has improved the reporting of thousands of medical experiments (Turner et al. 2012). It has been endorsed by over 600 journals (Moher et al. 2004), and it is supported by the Institute of Educational Sciences (Torgerson et al. 2005). CONSORT is the only guideline for reporting RCTs that has been developed with such rigour, and it has remained more prominent than any other guideline for over 15 years; for greatest

impact, then, any further reporting guidelines related to RCTs should be developed in collaboration with the CONSORT Group.

Limitations of previous reporting guidelines for social and psychological interventions

Researchers and journal editors in the social and behavioural sciences are generally aware of CONSORT, but often object that it is not fully appropriate for social and psychological interventions (Bonell et al. 2006; Davidson et al. 2003; Perry et al. 2010; Stinson et al. 2003). As a result, uptake of CONSORT guidelines in these disciplines is low. While some criticisms are due to inaccurate perceptions about common features of RCTs across disciplines, many relate to real limitations for social and psychological interventions (Mayo-Wilson 2007). For example, CONSORT is most relevant to RCTs in medical disciplines; it was developed by biostatisticians and medical researchers with minimal input from experts in other disciplines. Journal editors, as well as social and behavioural science researchers, believe there is a need to include appropriate stakeholders in developing a new, targeted guideline to improve uptake in their disciplines (Gill 2011; Torgerson et al. 2005). The CONSORT Group has produced *extensions* of the original CONSORT Statement relevant to social and psychological interventions, such as additional checklists for cluster (Campbell et al. 2004), non-pharmacological (Boutron et al. 2008a, b), pragmatic (Zwarenstein et al. 2008) and quality of life RCTs (Calvert et al. 2011). These extensions provide important insights, but complex social and psychological interventions, for example, include multiple, interacting components at several levels, with various outcomes. These RCTs require use of several extensions at once, creating a barrier to guideline uptake; increasing intervention complexity also gives rise to new issues that are not included in existing guidelines. Therefore, simply disseminating CONSORT guidelines as they stand is insufficient, as this would not address the need for editors and authors to “buy-in” to this process. To improve uptake in these disciplines, CONSORT guidelines need to be extended to specifically address the important features of social and psychological interventions.

Social and behavioural scientists have developed other reporting guidelines, including the Workgroup for Intervention Development and Evaluation Research (WIDER) Recommendations for behavioural change interventions (Abraham 2009; Michie et al. 2011), the American Educational Research Association’s (AERA) Standards for Reporting Research (AERA 2006), the REPOSE guidelines for primary research in education (Newman and Elbourne 2004), and the Journal Article Reporting Standards (JARS) of the American Psychological Association (APA 2008). Whilst they address issues not covered by the CONSORT Statement and its extensions, these guidelines (except for JARS; APA 2008) do not provide specific guidance for RCTs. Moreover, compared with the CONSORT Statement and its official extensions, guidelines in the social and behavioural sciences have not consistently followed optimal techniques for guideline development and dissemination that are recommended by international leaders in the advancement of reporting guidelines (Moher et al. 2010a, b), such as the use of systematic literature reviews

and formal consensus methods to select reporting standards (Grant et al. 2012). Researchers in public health, psychology, education, social work, and criminology have noted that these guidelines could be more ‘user-friendly’, and dissemination could benefit from up-to-date knowledge transfer techniques (Abraham 2009; Armstrong et al. 2008; Davidson et al. 2003; Naleppa and Cagle 2010; Perry and Johnson 2008; Stinson et al. 2003; Torgerson et al. 2005).

For example, JARS—a notable and valuable guideline for empirical psychological research—is endorsed by few journals outside the APA, whereas CONSORT is endorsed by hundreds of journals internationally. According to ISI Web of Knowledge and Google Scholar citations, JARS is cited approximately a dozen times annually, while CONSORT guidelines are cited hundreds of times per year. Moreover, the APA commissioned a select group of APA journal editors and reviewers to develop JARS, and the group based most of their work on existent CONSORT guidelines; by comparison, official CONSORT extensions have been developed using rigorous consensus methods, have involved various international stakeholders in guideline development and dissemination, and update content on the most recent scientific literature. Nonetheless, no current CONSORT guideline adequately addresses the unique features of social and psychological interventions. This new CONSORT extension will incorporate lessons from previous extensions, reporting guidelines, and the research literature to aid the critical appraisal, replication, and uptake of this research.

Aspects of internal validity

Internal validity is the extent to which the results of a study may be influenced by bias. Like other study designs, the validity of RCTs depends on high-quality execution. Poorly conducted RCTs can produce more biased results than well-conducted RCTs and well-conducted non-randomised studies (Pildal et al. 2007; Prescott et al. 1999). For example, evidence indicates that RCTs that do not adequately conceal the randomisation sequence can exaggerate effect estimates by up to 30 % (Schulz et al. 1995), while low-quality reports of these RCTs are associated with effect estimates exaggerated by up to 35 % (Moher et al. 1999). Social and psychological intervention RCTs are also susceptible to these risks of bias.

Some aspects of internal validity, although included in CONSORT, remain poorly reported—even in the least complex social and psychological intervention studies. Reports of RCTs should describe procedures for minimising selection bias, but reports often omit information about random sequence generation and allocation concealment (Ladd et al. 2010; Perry and Johnson 2008), and psychological journals report methods of sequence generation less frequently than medical journals (Stinson et al. 2003). A review of educational reports found no studies that adequately reported allocation concealment (Torgerson et al. 2005), and reports in criminology often lack information about randomisation procedures (Gill 2011; Perry et al. 2010). RCTs of social and psychological interventions may also use non-traditional randomisation techniques, such as stepped-wedge or natural allocation (Medical Research Council 2011), which need to be thoroughly described. In addition, reports of social and psychological intervention trials often fail to include details about trial registration, protocols, and

adverse events (Ladd et al. 2010; Perry and Johnson 2008), which may include important negative consequences at individual, familial, and community levels.

Other aspects of CONSORT may require greater emphasis or modification for RCTs of social and psychological interventions. In developing this CONSORT extension, we expect to identify new items and to adapt existing items that relate to the internal validity. These may include items discussed during the development of previous CONSORT extensions or other guidelines, as well as items suggested by participants in this project. For example, it may not be possible to blind participants and providers of interventions, but blinding of outcome assessors is often possible but rarely reported, and few studies explain if blinding was maintained or how lack of blinding was handled (Davidson et al. 2003; Ladd et al. 2010; Perry and Johnson 2008). In social and psychological intervention studies, outcome measures are often subjective, variables may relate to latent constructs, and information may come from multiple sources (e.g. participants, providers). While an issue in other areas of research, the influence on RCT results of the quality of subjective outcome measures in social and psychological intervention research has long been highlighted given their prevalence in social and psychological intervention research (Marshall et al. 2000). Descriptions of the validity, reliability, and psychometric properties of such measures are therefore particularly useful for social and psychological intervention trials, especially when they are not widely available nor discussed in the research literature (Campbell et al. 2004; Fraser et al. 2009). Moreover, multiple measures may be analysed in several ways, so authors need to transparently report which procedures were performed and to explain their rationale.

Aspects of external validity

External validity is the extent to which a study's results are applicable in other settings or populations. Currently, given that RCTs are primarily designed to increase the internal validity of study findings, the CONSORT Statement gives relatively little attention to external validity. While high internal validity is an important precondition for any discussion of an RCT's external validity, updating the CONSORT Statement to include more information about external validity is critical for the relevance and uptake of a CONSORT extension for social and psychological interventions. These interventions may be influenced by context, as different underlying social, institutional, psychological, and physical structures may yield different causal and probabilistic relationships between interventions and observed outcomes. Contextual information is necessary to compare the effectiveness of an interventions across time and place (Cartwright and Munro 2010). Lack of information relevant to external validity may prevent practitioners or policymakers from using evidence appropriately to inform decision-making, yet existing guidelines do not adequately explain how authors should describe (1) how interventions work, (2) for whom, and (3) under what conditions (Moore and Moore 2011).

Firstly, it is useful for authors to explain the key components of interventions, how those components could be delivered, and how they relate to the outcomes selected. At present, authors can follow current standards for reporting interventions without providing adequate details about complex interventions (Shepperd et al. 2009). Many reports contain neither sufficient information about the interventions tested nor reference treatment manuals (Glasziou et al. 2008). Providing logic models—as described in the Medical Research

Council (MRC) Framework for Complex Interventions (Craig et al. 2008)—or presenting theories of change can help elucidate links in causal chains that can be tested, identify important mediators and moderators, and facilitate syntheses in reviews (Ivers et al. 2012). Moreover, interventions are rarely implemented exactly as designed, and complex interventions may be designed to be implemented with some flexibility, in order to accommodate differences across participants (Hawe et al. 2004), so it is important to report how interventions were *actually delivered* by providers and *actually received* by participants (Hardeman et al. 2008). Particularly for social and psychological interventions, the integrity of implementing the intended functions and processes of the intervention are essential to understand (Hawe et al. 2004). As RCTs of a particular intervention can yield different relative effects depending on the nature of the control groups, information about delivery and uptake should be provided for *all* trial arms (McGrath et al. 2003).

Secondly, reports should describe recruitment processes and representativeness of samples. Participants in RCTs of social and psychological interventions are often recruited outside routine practice settings via processes that differ from routine services (AERA 2006). An intervention that works for one group of people may not work for people living in different cultures or physical spaces, or it may not work for people with slightly different problems and co-morbidities. Enrolling in an RCT can be a complex process that affects the measured and unmeasured characteristics of participants, and recruitment may differ from how users normally access interventions. Well-described RCT reports will include the characteristics of all participants (volunteers, those who enrolled, and those who completed) in sufficient detail for readers to assess the comparability of the study sample to populations and in everyday services (AERA 2006; APA 2008; Evans and Brown 2003)

Finally, given that these interventions often occur in social environments, reports should describe factors of the RCT context that are believed to support, attenuate, or frustrate observed effects (Moore 2002). Interventions may differ across groups of different social or socioeconomic positions, and equity considerations should be explicitly addressed (Tugwell et al. 2010; Welch et al. 2012). Several aspects of setting and implementation may be important to consider, such as administrative support, staff training and supervision, organisational resources, the wider service system, and concurrent political or social events (Bonell et al. 2012; Fixsen et al. 2005; Shepperd et al. 2009; Wang et al. 2006). Reporting process evaluations may help understand mechanisms and outcomes.

Developing a new CONSORT extension

This new reporting guideline for RCTs of social and psychological interventions will be an official extension of the CONSORT Statement. Optimally, it will help improve the reporting of these studies. Like other official CONSORT extensions (Boutron et al. 2008a, b; Campbell et al. 2004; Hopewell et al. 2008; Zwarenstein et al. 2008), this guideline will be integrated with the CONSORT Statement and previous extensions, and updates of the CONSORT Statement may incorporate references to this extension.

The project is being led by an international collaboration of researchers, methodologists, guideline developers, funders, service providers, journal editors, and consumer advocacy groups. We will be recruiting participants in a manner similar to other reporting guideline initiatives—identifying stakeholders through literature reviews, the project's International Advisory Group, and stakeholder-initiated interest

in the project (Michie et al. 2011; Schulz et al. 2010). We hope to recruit stakeholders with expertise from all related disciplines and regions of the world, including low- and middle-income countries. Methodologists will identify items that relate to known sources of bias, and they will identify items that facilitate systematic reviews and research synthesis. Funders will consider how the guideline can aid the assessment of grant applications for RCTs and methodological innovations in intervention evaluation. Practitioners will identify information that can aid decision-making. Journal editors will identify practical steps to implement the guideline and to ensure uptake.

We will use consensus techniques to reduce bias in group decision-making and to promote widespread guideline uptake and knowledge translation activities upon project completion (Murphy et al. 1998). Following rigorous reviews of existing guidelines and current reporting quality, we will conduct a Delphi process to identify a prioritised list of reporting items to consider for the extension. That is, we will invite a group of experts to answer questions about reporting items and to suggest further questions. We will circulate their feedback to the group and ask a second round of questions. The Delphi process will capture a variety of international perspectives and allow participants to share their views anonymously. Following the Delphi process, we will host a consensus meeting to review the findings and to generate a list of minimal reporting standards, mirroring the development of previous CONSORT guidelines (Boutron et al. 2008a, b; Schulz et al. 2010; Zwarenstein et al. 2008).

Together, participants in this process will create a checklist of reporting items and a flowchart for reporting social and psychological intervention RCTs. In addition, we will develop an Explanation and Elaboration (E&E) Document to explain the scientific rationale for each recommendation and to provide examples of clear reporting; a similar document was developed by the CONSORT group to help disseminate a better understanding for each included checklist item (Moher et al. 2010a, b). This document will help persuade editors, authors, and funders of the importance of the guideline. It will be a useful pedagogical tool, helping students and researchers understand the methods for conducting RCTs of social and psychological interventions, and it will help authors meet the guideline requirements (Moher et al. 2010a, b).

The success of this project depends on widespread involvement and agreement among key international stakeholders in research, policy, and practice. For example, previous developers have obtained guideline endorsement by journal editors who require authors and peer-reviewers to use the guideline during manuscript submission and who must enforce journal article word limits (Michie et al. 2009). Many journal editors have already agreed to participate, and we hope other researchers and stakeholders will volunteer their time and expertise.

Conclusion

Reporting guidelines help us use scarce resources efficiently and ethically. Randomised controlled trials are expensive, and the public have a right to expect returns on their investments through transparent, usable reports. When RCT reports cannot be used (for whatever reason), resources are wasted. Participants contribute their time and put themselves at risk of harm to generate evidence that will help others, and researchers

should disseminate that information effectively (Davidson et al. 2003). Policymakers benefit from research when developing effective, affordable standards of practice and choosing which programmes and services to fund. Administrators and managers are required to make contextually appropriate decisions. Transparent reporting of primary studies is essential for their inclusion in systematic reviews that inform these activities. For example, there is the need to determine if primary studies are comparable, examine biases within included studies, assess the generalisability of results, and implement effective interventions. Finally, we hope this guideline will reduce the effort and time required for authors to write reports of RCTs.

Randomised controlled trials are not the only valid method for evaluating interventions (Bonell et al. 2011), nor are they the only type of research that would benefit from better reporting (Goldbeck and Vitiello 2011). Colleagues have identified the importance of reporting standards for other types of research, including observational (von Elm et al. 2007), quasi-experimental (Des Jarlais et al. 2004), and qualitative studies (Tong et al. 2007). This guideline is the first step towards improving reports of many designs for evaluating social and psychological interventions, which we hope will be addressed by this and future projects. We invite stakeholders from disciplines that frequently research these interventions to join this important effort and participate in guideline development by visiting our website, where they can find more information about the project, updates on its progress, and sign up to be involved (<http://tinyurl.com/CONSORT-study>).

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